AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-14 | Entry window 14 October 2026 (full day) | 24 records screened in scope

24 RECORDS IN SCOPE (56 EXCLUDED)

9 OF 10 SUBTOPIC BANDS POPULATED

5 EFFECT ESTIMATES WITH A CI

2 TIER-A RECORDS

8 RECORDS CARRIED ON METADATA ALONE

Signal of the day

A national correction beat local fitting for PurpleAir in Philadelphia — RMSE 1.64 μg m^{-3} — and accuracy fell from R^2 0.68 to 0.57 within a year in the field.

Co-location across summer and spring plus a one-year re-evaluation. The Barkjohn (2021) US-wide correction applied to the mean of the two channels' `atm` outputs gave **RMSE 1.640** μg m^{-3}, **MBE 0.103**, **R^2 0.683** at both 1 h and 24 h, outperforming locally derived corrections. On-board vs weather-station RH changed R^2 by only 0.04. After a year, **R^2 0.567**. **Where it is weak:** R^2 at low temperate concentrations is range-limited, so part of the year-one drop may be a narrower concentration distribution rather than drift; the abstract does not separate the two. Barkjohn was derived on `cf_1`, so applying it to `atm` is off-label even if it works here. The operational reading is still clear: budget a scheduled re-co-location inside the first year.

What changed today

- **Two tier-A records, one of them calibration.** The Consensus sweep surfaced a mechanistic aerosol-liquid-water humidity correction (ACS ES&T Air, January 2026); the DOI gate showed it was already carried in the 2 August issue, so it is not repeated here — the one-record-one-issue rule held.
- **Smoke and the upper airway, again.** SEER county DLNM: any annual wildfire $PM_{2.5}$ exposure cRR **1.41** (1.24–1.60) for head and neck cancer over 0–10-year lags, larynx 1.13 per exceedance day — the second smoke–aerodigestive cancer signal in two days after the 13 October China fire-PM result.
- **Spatial information budget.** A 1 km CMAQ for Washington, D.C. finds NO_2's intra-urban coefficient of variation more than three times $PM_{2.5}$'s, and 4→1 km refinement moves $PM_{2.5}$ population-weighted exposure by only 1.5–3 %: dense $PM_{2.5}$ networks buy less spatial contrast than NO_2 networks would.
- **Indoor secondary particles.** Ventilated outdoor O_3 made indoor new particle formation VOC-limited; limonene drove growth at 20–100 nm h^{-1} — particles that $PM_{2.5}$-only indoor monitors do not see until they pass ∼300 nm.
- **Burden arithmetic.** Abu Dhabi attributable deaths rose from 231 to 1,249 while air quality held roughly steady — driven mainly by steeper concentration–response functions, not by exposure.

Window continuity

The 13 October issue (built earlier in this run) closed with `last_entry_date` at 13 October. This issue collects **14 October in full**, continuous with it, and brings the watch current. 15 October is not opened: this run began at 19:46 CDT on 15 October, before 22:00. The **W42 weekly** (11–17 October) is due with the Saturday 17 October issue.

Corpus analytics

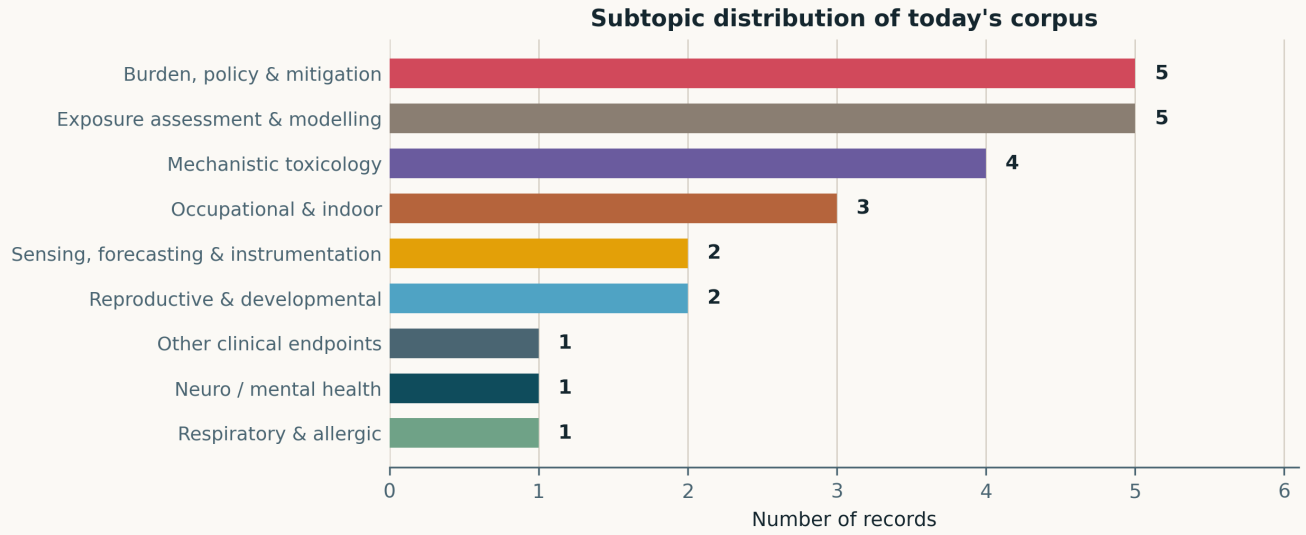


Fig. Subtopic assignment of the 24 in-scope records across nine bands. Exposure assessment and burden lead at five each; sensing (2) is small in count but carries one of the two tier-A records.

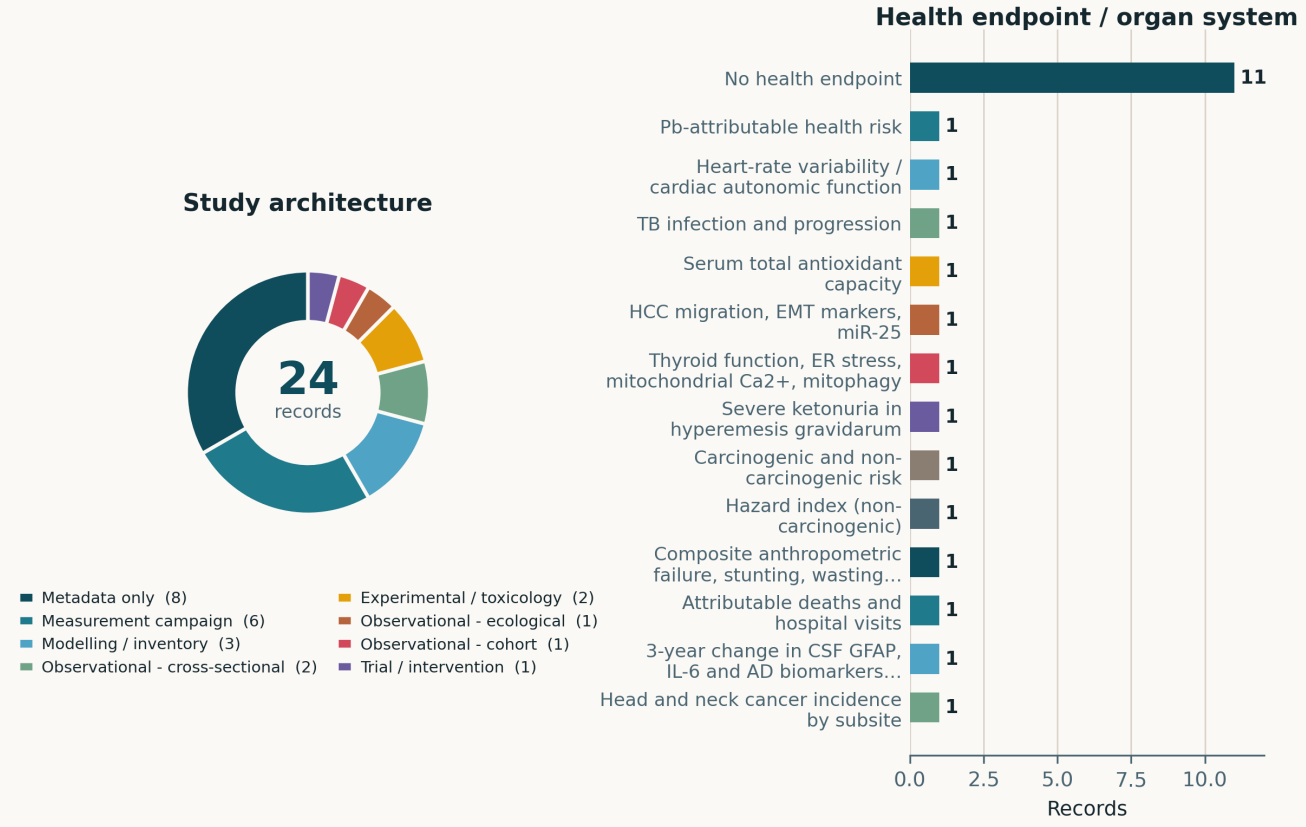


Fig. Left — architecture: eight metadata-only records (32 %), seven of them Elsevier deposits, the series' recurring blind spot. Right — health endpoints: cancer (head and neck, HCC), CSF biomarkers, undernutrition, hyperemesis, antioxidant capacity.

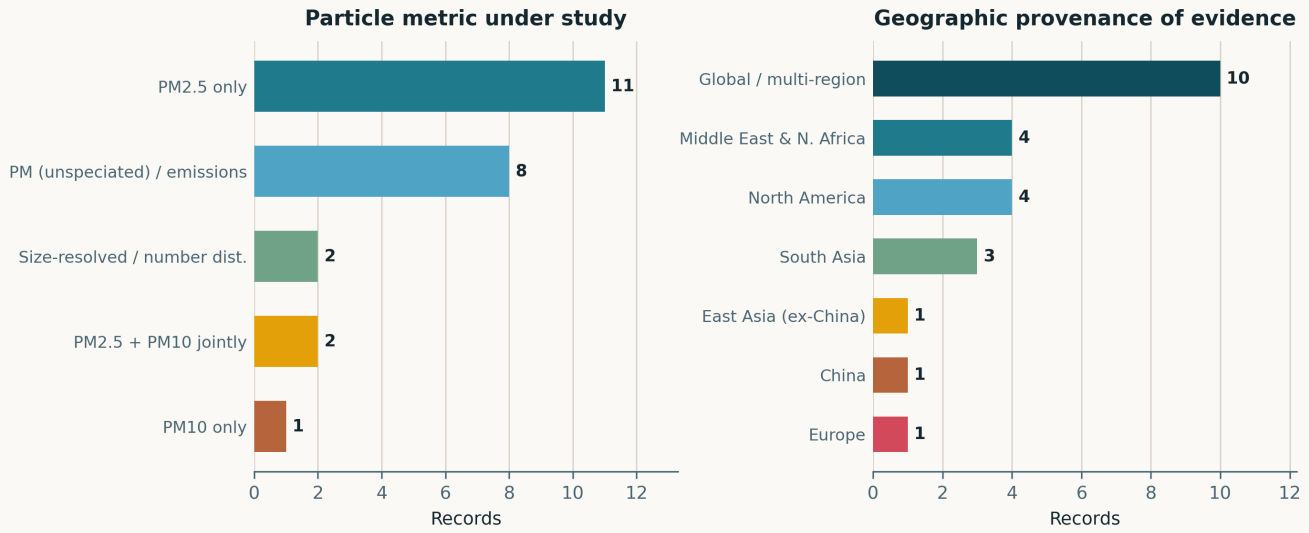


Fig. Left — particle metric: $PM_{2.5}$ leads; UFP, deposited dose, Pb, dust optics and indoor number distributions appear once each. Right — geography: United States leads (4), with India, Pakistan, Iran, UAE, Spain, Morocco and Turkiye.

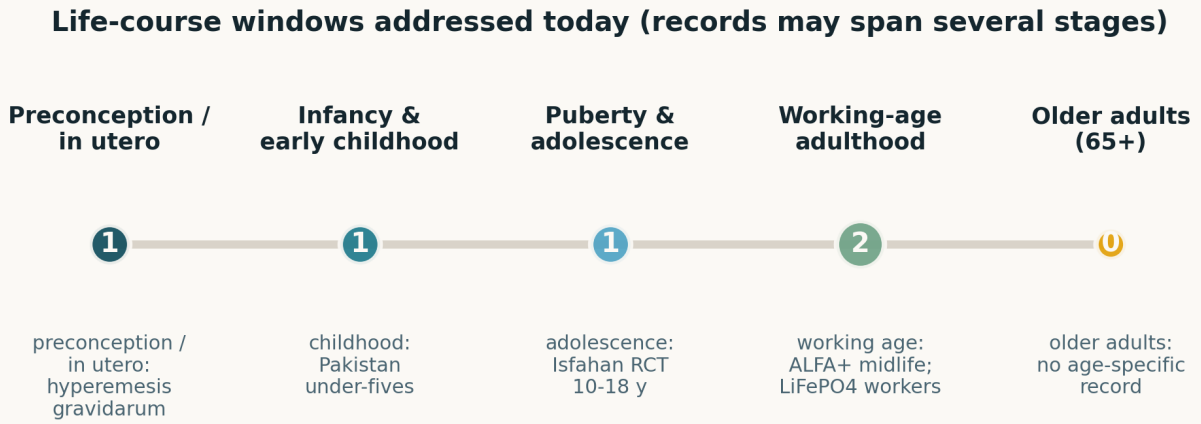


Fig. Life-course windows: pregnancy (hyperemesis), under-fives (Pakistan), adolescents (Isfahan trial), working age (ALFA+ midlife, cathode workers). No older-adult-specific record.

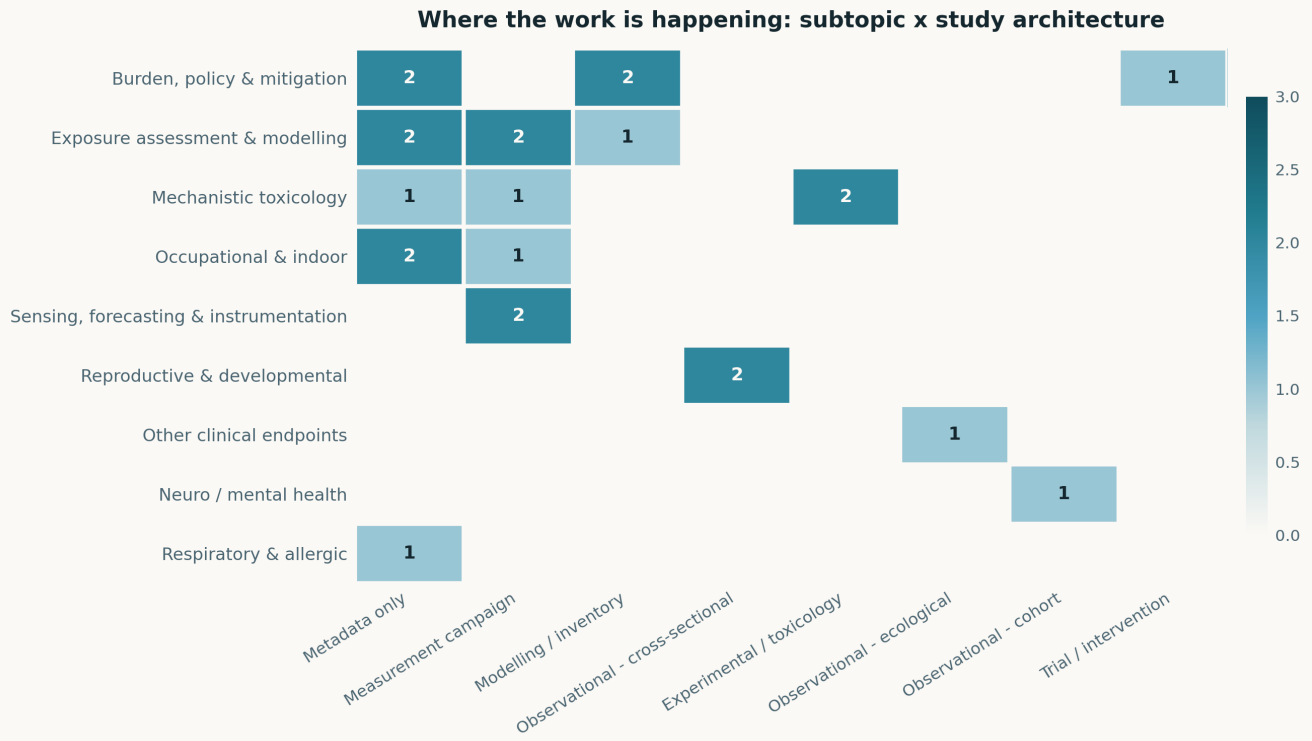


Fig. Subtopic × study architecture. The metadata-only column spans five bands; the sensing row is entirely field measurement.

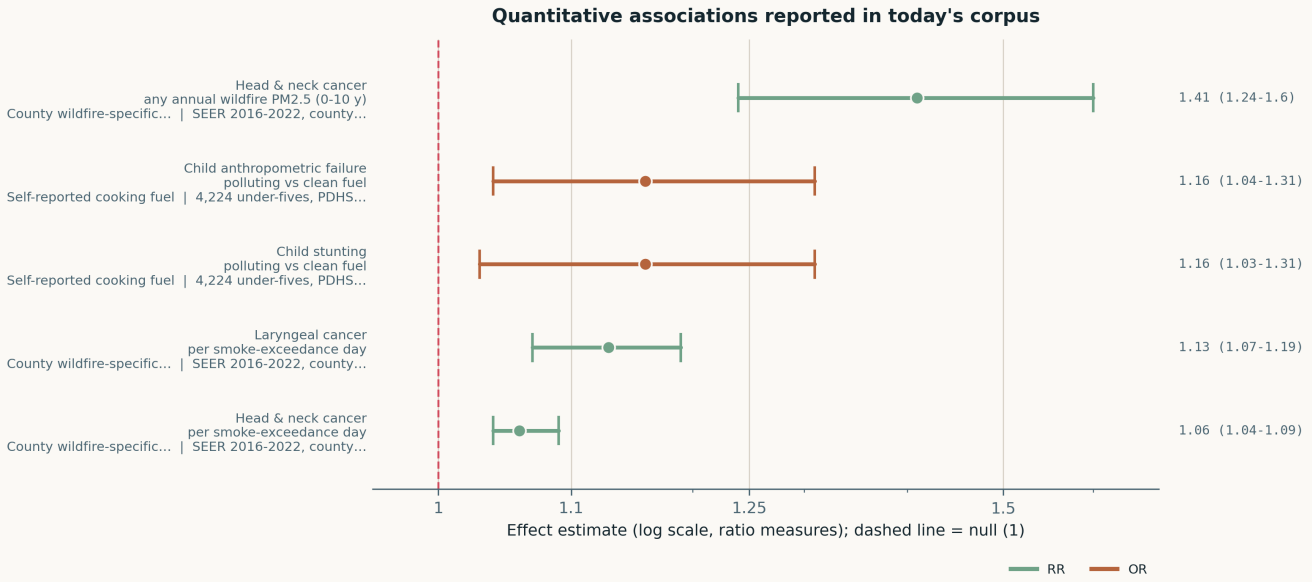


Fig. Five ratio estimates with 95 % CIs from two studies. The three head-and-neck cancer estimates are cumulative RRs from one county-level DLNM on three different exposure definitions — not independent. The two Pakistan ORs compare polluting with clean cooking fuel, a wealth-confounded proxy with no measured PM. None of the five is per unit concentration.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 1 record

ALFA+ TRAP CSF GFAP 2026 Alzheimers Dement

Air pollution is associated with sex-specific longitudinal changes in astrocytic and neuroinflammatory processes linked to cognitive decline in Alzheimer's disease at-risk individuals

226 cognitively unimpaired middle-aged adults at elevated AD risk with baseline and 3-year CSF. Higher traffic-related exposure associated with longitudinal increases in GFAP and IL-6, mainly in women; GFAP partially mediated associations with worsening executive and attentional performance. Limitation: no effect sizes in the abstract; n=226 split by sex across four correlated pollutants and many biomarkers is a multiplicity problem, and single-address LUR estimates ignore time-activity. Relevance: moderate - CSF longitudinal biology is rare, but this is hypothesis-generating at this n.

doi: 10.1002/alz.71837

Respiratory & allergic — 1 record

$PM_{2.5}$ constituents TB contacts 2026 Ann Am Thorac Soc

Differential effects of $PM_{2.5}$ constituents on tuberculosis infection and disease progression in 40,379 student close contacts

METADATA ONLY. No abstract at harvest. Title indicates a constituent-resolved analysis of $PM_{2.5}$ and latent infection and progression among 40,379 student close contacts of TB cases. Recorded for coverage; no estimate inferred. Relevance: moderate - a large contact cohort is an unusually clean design for an infection endpoint.

doi: 10.1093/annalsats/aaoag325

Reproductive & developmental — 2 records

Pakistan cooking fuel undernutrition 2026 Int J Environ Health Res

Association between household fuel use and childhood undernutrition in Pakistan: evidence from a National survey

4,224 under-fives; 45.4% with anthropometric failure, 37.7% stunted. Polluting vs clean fuels: CIAF OR 1.16 (1.04-1.31), stunting 1.16 (1.03-1.31), wasting 1.13 (1.00-1.29), underweight 1.17 (1.04-1.32); dung strongest. Limitation: fuel type is a wealth marker as much as an exposure, and with no measured PM the residual socioeconomic confounding is plausibly the size of the effect; cross-sectional. Relevance: moderate - the standard HAP proxy, and a reminder of why personal-monitor substudies inside DHS-type surveys are needed.

doi: 10.1080/09603123.2026.2749055

Hyperemesis gravidarum PM_{10} temperature 2026 Front Public Health

Seasonal ambient temperature and particulate matter as discriminators of hyperemesis gravidarum severity: a retrospective comparative observational study

211 HG patients: temperature (AUC 0.799) and PM_{10} (AUC 0.754) discriminated severe from non-severe ketonuria; SO_2 did not. The PM_{10} association attenuated after adjustment for temperature. Limitation: hospital series with no control group, ecological exposure, and PM_{10} and temperature are seasonally collinear - the attenuation says PM_{10} was largely a season marker. Relevance: low.

doi: 10.3389/fpubh.2026.1957434

Mechanistic toxicology 4 records

Houston UFP PAH deposition 2026 Air Qual Atmos Health

PAH-associated health risk from ultrafine particles: an urban field study

Three Houston sites: UFP respiratory deposition measured directly with the Mobile Aerosol Lung Deposition Apparatus, combined with size-segregated PAH analysis to estimate deposited PAH mass. Both carcinogenic and non-carcinogenic risks were within acceptable levels. Limitation: no numbers in the abstract; PAH risk is one narrow axis of UFP toxicity, so 'acceptable' does not translate to UFP being benign. Relevance: moderate - measured deposited dose rather than ambient concentration is the right exposure metric and is rarely available.

doi: 10.1007/s11869-026-02118-x

$PM_{2.5}$ thyroid vitamin C rats 2026 Ecotoxicol Environ Saf

Vitamin C attenuates $PM_{2.5}$-induced thyroid dysfunction and mitophagy dysregulation by rebalancing ER-mitochondrial Ca2+ dynamics

Female rats and FRTL5 thyrocytes: $PM_{2.5}$ impaired thyroid function and histology, triggered ER stress, remodelled MAM proteins (IP3R1, GRP75, VDAC1), caused mitochondrial Ca2+ overload, ROS, depolarisation, PINK1/Parkin changes, GPX4 loss and Fe2+ accumulation; 4-PBA, 2-APB, Mdivi-1 and vitamin C each mitigated damage. Limitation: doses not given in the abstract and likely supra-environmental. Relevance: low - mechanism for a thyroid signal that the epidemiology (13 Oct fire-PM thyroid cancer) has not yet tied to function.

doi: 10.1016/j.ecoenv.2026.120920

$PM_{2.5}$ HCC miR-25 2026 Transl Cancer Res

$PM_{2.5}$ promotes migration of hepatocellular carcinoma by targeting miR-25

$PM_{2.5}$ lowered E-cadherin, raised vimentin and miR-25 and increased MHCC97H migration; miR-25 mimics amplified and inhibitors attenuated the effect; mouse models showed enhanced tumour growth and invasion. Limitation: single cell line, unstated doses, and a single-miRNA mechanism for a pleiotropic exposure. Relevance: low.

doi: 10.21037/tcr-2026-0728

Mouth-throat replica deposition 2026 J Aerosol Sci

The influence of model selection and flow rate on aerosol transport in replica models of the mouth and throat: An experimental study

METADATA ONLY. No abstract at harvest. Title indicates an experimental comparison of mouth-throat replica geometries and flow rates for aerosol transport. Recorded for coverage. Relevance: low-moderate - dosimetry inputs.

doi: 10.1016/j.jaerosci.2026.106906

Exposure assessment & modelling 5 records

DC 1 km WRF-SMOKE-CMAQ 2026 ACS EST Air

Neighborhood-Scale O_3, NO_2, and $PM_{2.5}$ in Washington, D.C. from a 1 x 1 km2 WRF-SMOKE-CMAQ Framework

1 km CMAQ for 2017: hourly O_3 r up to 0.7; NO_2 biased -14 to -29% from unresolved near-road gradients. NO_2's coefficient of variation exceeded O_3's and was more than three times $PM_{2.5}$'s; urban-core O_3 was 4-11% lower from titration. Moving from 4 to 1 km changed population-weighted exposure by only -1.5 to -3% for $PM_{2.5}$. Limitation: evaluation against a handful of AQS sites in one city cannot test intra-urban gradients - the very thing 1 km is meant to add. Relevance: high - quantifies that $PM_{2.5}$ is the spatially smoothest pollutant, so dense $PM_{2.5}$ networks buy less spatial information than NO_2 networks would.

doi: 10.1021/acsestair.6c00155

Lucknow Holika Dahan $PM_{2.5}$ 2026 Environ Monit Assess

Impact of Holika Dahan on ambient $PM_{2.5}$ and associated health risks in an urban environment of Lucknow, India

Pre-, during- and post-Holika Dahan nights at a residential and a commercial site: $PM_{2.5}$ rose 2.0x and 1.4x; burning tracers 1.2-2.7x; PCA/HCA identified festival biomass burning as the dominant source; AERMOD placed hotspots out to 1.2-1.5 km. Hazard indices stayed below USEPA limits. Limitation: two sites over three nights, and a chronic-dose hazard index applied to a single-night event is the wrong risk metric. Relevance: moderate - a short, intense, spatially bounded event of exactly the kind a dense network resolves and two filter sites do not.

doi: 10.1007/s10661-026-16041-2

Delhi-NCR COVID transitions 2026 Sci Prog

Time series evidence on air quality transitions under urban shock during COVID-19 shutdown in Delhi-NCR, India

Ten regulatory stations across pre-, during- and post-COVID periods analysed with ARIMA, PCA, change-point detection and recurrence quantification: significant reductions and altered temporal dynamics during lockdown, with meteorology and secondary formation relatively more important. Limitation: no deweathering is described, so lockdown effects are confounded with seasonal meteorology; the lockdown literature on Delhi is saturated. Relevance: low.

doi: 10.1177/00368504261498073

China 20 cities building green $PM_{2.5}$ 2026 Environ Pollut

City-size-dependent interactive effects of building and green-space on $PM_{2.5}$ pollution: Evidence from 20 representative Chinese cities

METADATA ONLY. No abstract at harvest. Title indicates interactions of building form and green space with $PM_{2.5}$ that depend on city size. Recorded for coverage. Relevance: low-moderate.

doi: 10.1016/j.envpol.2026.129344

Gray-green infrastructure $PM_{2.5}$ MSPA 2026 Build Environ

Spatially Explicit Morphological Regulation of Gray-Green Infrastructure for $PM_{2.5}$ Mitigation in High-Density Cities: An Interpretable MSPA-Machine Learning Framework

METADATA ONLY. No abstract at harvest. Title indicates morphological spatial pattern analysis plus interpretable ML relating gray-green infrastructure to $PM_{2.5}$ in dense cities. Recorded for coverage. Relevance: low.

doi: 10.1016/j.buildenv.2026.115406

Sensing, forecasting & instrumentation 2 records

Sylvester PurpleAir Philadelphia 2026 ACS EST Air

Validation of PurpleAir $PM_{2.5}$ Sensors in Temperate Urban Environments

Co-location in Philadelphia across summer and spring plus a one-year re-evaluation. Best performance came from the nationwide Barkjohn (2021) correction applied to the mean of the two channels' atm outputs: RMSE 1.64 $\mu g\,m^{-3}$, MBE 0.10 $\mu g\,m^{-3}$, R2 0.683 at 1 h and 24 h, ahead of locally derived corrections. RH source (on-board vs weather station) changed R2 by only 0.04. After a year in the field R2 fell to 0.567. Limitation: R2 0.68 at low temperate concentrations means RMSE is the honest metric and the R2 drop partly reflects a narrower concentration range in the later period, not drift alone - the abstract does not separate them. Applying Barkjohn, derived on cf_1, to atm outputs is off-label. Relevance: high - direct evidence that a national correction can beat local fitting, and that year-one degradation needs scheduled re-co-location.

doi: 10.1021/acsestair.5c00443

Saharan fresh dust optics 2026 Atmos Chem Phys

Optical properties and shape-dependent retrievals of the imaginary part of the refractive index of freshly emitted Saharan dust

In-situ optics during active emission events (Sept 2019): SSA for $PM_{2.5}$ (PM_{10}) 0.95 (0.94) at 370 nm and 0.97 (0.96) at 660 nm; AAE to 2.5 (2.0) from iron oxides; MAE fell from 0.30 to 0.15 m2/g at 370 nm with size. The imaginary index was retrieved under prescribed shape and density. Limitation: the retrieval depends on the shape assumption it is titled after, and a single campaign. Relevance: moderate - mass scattering efficiency shifting with size is the physics behind optical sensors' coarse-dust under-response; these values are usable priors for a dust correction.

doi: 10.5194/acp-26-14531-2026

Occupational & indoor 3 records

Indoor NPF from outdoor O_3 2026 Environ Sci Technol

Elevated Outdoor O_3 Can Sustain New Particle Formation and Growth Indoors

In an authentic office in a high-O_3 region, mechanical ventilation supplied enough O_3 that indoor new particle formation was VOC-limited, not O_3-limited; limonene addition drove growth at 20-100 nm/h, one to two orders above outdoor rates, with meaningful mass from grown particles. Limitation: one building, injected limonene rather than real product use, and no occupancy. Relevance: high - indoor UFP and secondary mass from cleaning products are invisible to $PM_{2.5}$-only indoor sensors until particles grow past ~300 nm.

doi: 10.1021/acs.est.6c07592

LiFePO4 workers cardiac autonomic 2026 Occup Environ Med

Association between occupational LiFePO4 exposure and cardiac autonomic function: a cross-sectional study of cathode material production workers

METADATA ONLY. No abstract at harvest. Title indicates a cross-sectional study of battery-cathode workers' LiFePO4 particle exposure and cardiac autonomic function. Recorded for coverage. Relevance: moderate - battery-material dust is an emerging occupational particle class with little exposure data.

doi: 10.1136/oemed-2025-110695

Office air purifier field CADR 2026 Build Environ

Air purifier performance in an occupied office: Field CADR and effectiveness from a reconstructed device-off condition

METADATA ONLY. No abstract at harvest. Title indicates field estimation of purifier clean-air delivery rate in an occupied office using a reconstructed device-off counterfactual. Recorded for coverage. Relevance: high if the counterfactual is sensor-derived - field CADR is how purifier-trial exposure contrasts should be verified.

doi: 10.1016/j.buildenv.2026.115390

Burden, policy & mitigation — 5 records

Global heatwave-fire smoke coexposure 2026 Sci Adv

Rising global population coexposure to heatwaves and landscape fire-sourced air pollution

Global 2000-2019: 55.9% of the population lived where heatwave-smoke coexposure increased; 916 million people/year had at least one coexposure day, rising by 287 million per decade; mean coexposure days per person rose from 1.0 to 1.7. Of the 5.5 billion person-day/decade increase, 70.5% was intensity and 29.5% population growth; low-HDI countries carried the most. Limitation: the fire-PM product is model-derived with no stated threshold sensitivity, and 'coexposure day' is a binary that discards concentration. Relevance: moderate - joint heat-smoke days are also when optical sensors are most biased (low RH, fresh smoke), so the exposure most in need of measurement is the hardest to measure cheaply.

doi: 10.1126/sciadv.aea9749

Abu Dhabi PM burden 2008-2022 2026 Risk Anal

Ambient Air Pollution Health Impacts in High-Income Countries, Part II: A Case Study in the United Arab Emirates

Attributable deaths from anthropogenic PM and ozone in Abu Dhabi rose from 231 (90% UI 59-427) in 2008 to 1,249 (507-2,169) in 2019-2022, despite air quality holding roughly steady while population doubled. The increase reflects population growth, better health records and, mainly, steeper concentration-response functions. Limitation: a fivefold change driven largely by swapping CRFs is a change in assumptions, not in burden; the 'anthropogenic' split in a dust-dominated desert regime is itself model-dependent. Relevance: moderate - shows how far CRF choice swamps measured trends, which bounds what better monitoring alone can change.

doi: 10.1111/risa.70368

Isfahan synbiotic omega-3 RCT 2026 J Res Med Sci

Effects of synbiotic versus omega-3 supplementation on total antioxidant capacity in adolescents exposed to ambient air pollution: A randomized double-blind controlled trial

70 healthy adolescents aged 10-18, 8 weeks synbiotic (n=38) vs omega-3 (n=32). TAC rose within both arms (839.8 to 903.4 and 898.6 to 953.3 umol/L) with no between-group difference (p=0.75). Limitation: active-comparator trial with no placebo arm, so within-arm rises cannot be separated from regression to the mean or seasonal change; exposure is not measured, so 'exposed to air pollution' is residence in Isfahan. Relevance: low.

doi: 10.4103/jrms.jrms_753_25

Global atmospheric Pb 2000-2019 2026 Environ Pollut

Global modelling of atmospheric lead (Pb) concentrations and health risks during 2000-2019

METADATA ONLY. No abstract at harvest. Title indicates global modelling of atmospheric Pb and associated risk over two decades. Recorded for coverage. Relevance: low.

doi: 10.1016/j.envpol.2026.129346

East Asia $PM_{2.5}$ O_3 warming levels 2026 Environ Pollut

Uneven Future $PM_{2.5}$ and O_3 responses to Emission Reductions under Different Warming Levels in East Asia

METADATA ONLY. No abstract at harvest. Title indicates climate-penalty-modulated $PM_{2.5}$ and O_3 responses to emission cuts in East Asia. Recorded for coverage. Relevance: low.

doi: 10.1016/j.envpol.2026.129354

Other clinical endpoints — 1 record

SEER wildfire $PM_{2.5}$ head and neck cancer 2026 Laryngoscope

Wildfire Air Pollution Exposure and Head and Neck Cancer Incidence

SEER 2016-2022, county-level DLNM with 0-10-year lags, adjusted for tobacco, alcohol and demographics with county random effects. Cumulative RR per additional smoke-exceedance day 1.06 (1.04-1.09); any annual exposure 1.41 (1.24-1.60); per percentage-point of days >5 $\mu g\,m^{-3}$ 1.23 (1.13-1.33). Larynx largest (1.13, 1.07-1.19 per day), oropharynx 1.09. Limitation: ecological county-level exposure and covariates, a ten-year lag fitted on seven years of incidence, and wildfire-exposed western counties differ in screening and HPV epidemiology; the p16 stratification is the right control but its result is truncated in the abstract. Relevance: moderate-high - a second smoke-cancer signal this week (cf. 13 Oct China), both favouring upper aerodigestive sites.

doi: 10.1002/lary.70874

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
ALFA+ TRAP CSF GFAP	Air pollution is associated with sex-specific longitudinal changes in astrocytic and neuroinflammatory processes linked to cognitive decline in Alzheimer's disease at-risk individuals	*Alzheimers Dement*	Prospective cohort	LUR NO2, PM_{10}, $PM_{2.5}$, $PM_{2.5}$ absorbance	Spain (Barcelona, ALFA+)
Reproductive & developmental					
Hyperemesis gravidarum PM_{10} temperature	Seasonal ambient temperature and particulate matter as discriminators of hyperemesis gravidarum severity: a retrospective comparative observational study	*Front Public Health*	Observational - cross-sectional	Ambient PM_{10}, SO2 and temperature (regional)	Turkiye
Pakistan cooking fuel undernutrition	Association between household fuel use and childhood undernutrition in Pakistan: evidence from a National survey	*Int J Environ Health Res*	Observational - cross-sectional	Household air pollution proxied by self-reported primary cooking fuel	Pakistan (national, PDHS 2017-18)
Respiratory & allergic					
$PM_{2.5}$ constituents TB contacts	Differential effects of $PM_{2.5}$ constituents on tuberculosis infection and disease progression in 40,379 student close contacts	*Ann Am Thorac Soc*	Metadata only (no abstract)	$PM_{2.5}$ constituents	Not stated
Mechanistic toxicology					
Houston UFP PAH deposition	PAH-associated health risk from ultrafine particles: an urban field study	*Air Qual Atmos Health*	Field measurement	UFP respiratory deposition (MALDA) and size-segregated UFP-bound PAHs	United States (Greater Houston)
Mouth-throat replica deposition	The influence of model selection and flow rate on aerosol transport in replica models of the mouth and throat: An experimental study	*J Aerosol Sci*	Metadata only (no abstract)	Aerosol deposition in mouth-throat replicas	Laboratory
$PM_{2.5}$ HCC miR-25	$PM_{2.5}$ promotes migration of hepatocellular carcinoma by targeting miR-25	*Transl Cancer Res*	Animal + in vitro	$PM_{2.5}$ exposure of MHCC97H cells and mouse tumour models	Laboratory
$PM_{2.5}$ thyroid vitamin C rats	Vitamin C attenuates $PM_{2.5}$-induced thyroid dysfunction and mitophagy dysregulation by rebalancing ER-mitochondrial Ca2+ dynamics	*Ecotoxicol Environ Saf*	Animal + in vitro	$PM_{2.5}$ instillation (rats) and FRTL5 thyrocyte exposure	Laboratory
Occupational & indoor					
Indoor NPF from outdoor O3	Elevated Outdoor O3 Can Sustain New Particle Formation and Growth Indoors	*Environ Sci Technol*	Field measurement	Indoor particle number size distributions; ventilated O3; limonene addition	Not stated (aged unoccupied office, high-O3 region)
LiFePO4 workers cardiac autonomic	Association between occupational LiFePO4 exposure and cardiac autonomic function: a cross-sectional study of cathode material production workers	*Occup Environ Med*	Metadata only (no abstract)	Occupational LiFePO4 particle exposure	Not stated
Office air purifier field CADR	Air purifier performance in an occupied office: Field CADR and effectiveness from a reconstructed device-off condition	*Build Environ*	Metadata only (no abstract)	Indoor PM with portable air cleaner (field CADR)	Not stated
Sensing, forecasting & instrumentation					
Saharan fresh dust optics	Optical properties and shape-dependent retrievals of the imaginary part of the refractive index of freshly emitted Saharan dust	*Atmos Chem Phys*	Field measurement	Size-resolved SSA, AAE, asymmetry, MAE/MSE; retrieved imaginary refractive index	Morocco (Sahara)
Sylvester PurpleAir Philadelphia	Validation of PurpleAir $PM_{2.5}$ Sensors in Temperate Urban Environments	*ACS EST Air*	Field measurement	PurpleAir PA-II $PM_{2.5}$ (cf_1 vs atm outputs) co-located with reference; RH from PA and weather station	United States (Philadelphia)

First author	Focus	Journal	Design	Metric	Region
Exposure assessment & modelling					
China 20 cities building green $PM_{2.5}$	City-size-dependent interactive effects of building and green-space on $PM_{2.5}$ pollution: Evidence from 20 representative Chinese cities	*Environ Pollut*	Metadata only (no abstract)	$PM_{2.5}$ vs urban morphology	China (20 cities)
DC 1 km WRF-SMOKE-CMAQ	Neighborhood-Scale O3, NO2, and $PM_{2.5}$ in Washington, D.C. from a 1 x 1 km2 WRF-SMOKE-CMAQ Framework	*ACS EST Air*	Chemical transport model	Hourly O3, NO2, $PM_{2.5}$ at 1 km (2017)	United States (Washington, D.C.)
Delhi-NCR COVID transitions	Time series evidence on air quality transitions under urban shock during COVID-19 shutdown in Delhi-NCR, India	*Sci Prog*	Monitoring-record analysis	Daily $PM_{2.5}$, PM_{10}, NOx, SO2, CO, O3 at 10 CAAQMS stations	India (Delhi-NCR)
Gray-green infrastructure $PM_{2.5}$ MSPA	Spatially Explicit Morphological Regulation of Gray-Green Infrastructure for $PM_{2.5}$ Mitigation in High-Density Cities: An Interpretable MSPA-Machine Learning Framework	*Build Environ*	Metadata only (no abstract)	$PM_{2.5}$ vs green-space morphology (MSPA)	Not stated
Lucknow Holika Dahan $PM_{2.5}$	Impact of Holika Dahan on ambient $PM_{2.5}$ and associated health risks in an urban environment of Lucknow, India	*Environ Monit Assess*	Field measurement	Night-time $PM_{2.5}$ mass and chemistry (tracers, ATR-FTIR), HYSPLIT, AERMOD	India (Lucknow)
Burden, policy & mitigation					
Abu Dhabi PM burden 2008-2022	Ambient Air Pollution Health Impacts in High-Income Countries, Part II: A Case Study in the United Arab Emirates	*Risk Anal*	Health impact model	Anthropogenic PM and ozone, 2019-2022 vs 2008	United Arab Emirates (Abu Dhabi)
East Asia $PM_{2.5}$ O3 warming levels	Uneven Future $PM_{2.5}$ and O3 responses to Emission Reductions under Different Warming Levels in East Asia	*Environ Pollut*	Metadata only (no abstract)	Projected $PM_{2.5}$ and O3 under warming levels	East Asia
Global atmospheric Pb 2000-2019	Global modelling of atmospheric lead (Pb) concentrations and health risks during 2000-2019	*Environ Pollut*	Metadata only (no abstract)	Modelled airborne (particle-bound) Pb	Global / multi-region
Global heatwave-fire smoke coexposure	Rising global population coexposure to heatwaves and landscape fire-sourced air pollution	*Sci Adv*	Exposure model (secondary data)	Landscape fire-sourced $PM_{2.5}$ (modelled) coincident with heatwave days	Global / multi-region
Isfahan synbiotic omega-3 RCT	Effects of synbiotic versus omega-3 supplementation on total antioxidant capacity in adolescents exposed to ambient air pollution: A randomized double-blind controlled trial	*J Res Med Sci*	Randomised controlled trial	Ambient air pollution (not measured at individual level)	Iran (Isfahan)
Other clinical endpoints					
SEER wildfire $PM_{2.5}$ head and neck cancer	Wildfire Air Pollution Exposure and Head and Neck Cancer Incidence	*Laryngoscope*	Ecological / registry analysis	County-level wildfire-specific $PM_{2.5}$ (exceedance days, any exposure, % days > 5 $\mu g/m^3$), 0-10-y lags	United States (SEER counties)

Provenance and method

Entry window **14 October 2026**, single day. Harvester legs: **PubMed** health **26**, sensing **4**; **Europe PMC 0** (index lag; the 14 October creation date had not populated by harvest); **Crossref by ISSN 48**; **arXiv** no entry dated 14 October; **OpenAlex** HTTP 429. 75 unique, 0 already carried. **PubMed connector** 18 PMIDs dated 14 October, 1 merged, 5 appended — **80** candidates screened, **24 admitted**, **56 rejected** with logged reasons (one, the GeoHealth ammonia model, a PubMed copy of a 13 October Crossref record). **Consensus** calibration sweep: 20 hits, **0 new** (one apparent new hit, ACS ES&T Air 2026, was caught by the DOI gate as shipped on 2 August). Abstract retry recovered 15 of 24; the PurpleAir validation abstract came from Consensus; **8 records carried on metadata alone**. **Scholar Gateway** still `ACCESS_DENIED`. Abstracts remain the copyright of their publishers.